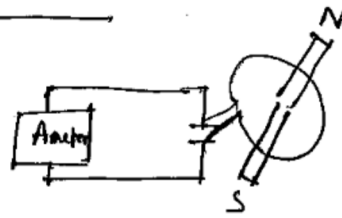


Electromagnetic Induction - Chapter 30,

R.H.W 10th Edition.

Faraday's Law and Lenz's Law



(i) If we close switch 'S' to turn on a current in the right-hand loop, the meter suddenly and briefly registers a current.

(i) A current appears only if there is relative motion between the loop and the magnet.

(ii) If we then open the switch, another sudden and brief induced current appears in the left-hand loop, but in opposite direction.

(ii) Faster motion produces a greater current.

(iii) If moving the magnet's north pole toward the loop causes, say, clockwise current, then moving the north pole away causes counterclockwise current.

Faraday's Law of Induction

An emf is induced in the loop at the left in above figures when the number of magnetic field lines that pass through the loop is changing.

A Quantitative Treatment

here we need to define a magnetic flux. A loop enclosing an area A is placed in a magnetic field $\vec{B}$. Then the magnetic flux through the loop is $\phi_B = \int \vec{B} \cdot d\vec{A}$ (magnetic flux through area A).

Now law in terms of magnetic flux: The magnitude of the emf $\mathcal{E}$ induced in a conducting loop is equal to the rate at which the magnetic flux ϕ_B through that loop changes with time.

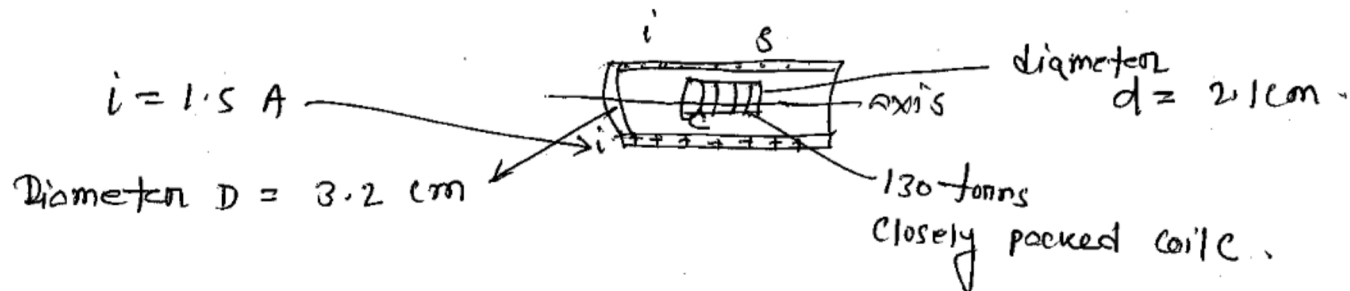
Faraday's law is formally written as;

$$\mathcal{E} = - \frac{d\Phi_B}{dt} \quad (\text{Faraday's law})$$

If we change the magnetic flux through a coil of N turns, an induced emf appears in every turn and the total emf induced the coil is the sum of these individual induced emfs.

$$\mathcal{E} = -N \frac{d\Phi_B}{dt} \quad (\text{coil of } N \text{ turns})$$

Sample Problem 30.01 induced emf in coil due to a solenoid.



The current in the solenoid is reduced to zero at a steady rate in 25 ms. What is the magnitude of the emf that is induced in coil C while the current in the solenoid is changing?

→ Faraday law → $\mathcal{E} = -N \frac{d\Phi_B}{dt}$

The final flux $\Phi_{B,f} = 0$ (current in the solenoid zero)

$$\Phi_{B,i} = ? \rightarrow \text{Area} = \frac{1}{4} \pi d^2$$

$$\Phi_{B,i} = BA = (\mu_0 i n) A = 1.44 \times 10^{-5} \text{ Wb}$$

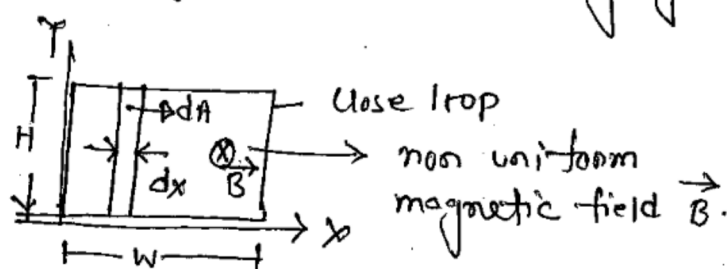
$$\frac{d\Phi_B}{dt} = \frac{\Delta\Phi_B}{\Delta t} = \frac{\Phi_{B,f} - \Phi_{B,i}}{\Delta t} = -5.76 \times 10^{-4} \text{ V}$$

$$\mathcal{E} = N \frac{d\Phi_B}{dt} = (130 \text{ turns}) (5.76 \times 10^{-4} \text{ V}) = 75 \text{ mV}$$

Lenz's Law An induced current has a direction such that the magnetic field due to the current opposes the change in the magnetic flux that induces the current.

- (i) Opposition to pole Movement
- (ii) Opposition to flux change.

Sample problem 30.03 Induced emf due to a changing nonuniform B field.



The field magnitude is given by $B = 4t^2x^2$, with B in teslas, t in seconds and x in meters.

$w = 3.0 \text{ m}$ height $H = 2.0 \text{ m}$.

Q → What are the magnitude and direction of the induced emf $\mathcal{E}$ around the loop at $t = 0.10 \text{ s}$?

→ $\vec{B}$ changing means Φ_B changing

→ from Faraday's law $\mathcal{E} = -\frac{d\Phi_B}{dt}$

$$\Phi_B = \int \vec{B} \cdot d\vec{A} = \int B dA = \int B H dx = \int 4t^2x^2 H dx$$

$$\Phi_B = 4t^2 H \int_0^{3.0} x^2 dx = 4t^2 H \left[\frac{x^3}{3} \right]_0^{3.0} = 72t^2 \quad [H = 2.0 \text{ m}]$$

$$\mathcal{E} = \frac{d\Phi_B}{dt} = \frac{d}{dt} (72t^2) = 144t \quad (\text{volt})$$

at $t = 0.10 \text{ s}$, $\mathcal{E} = (144 \text{ V/s})(0.10 \text{ s}) \approx 14 \text{ V}$

($\mathcal{E} = 14 \text{ V}$ is answer)

A reformulation of Faraday's Law

The work W done on it in one revolution by the induced electric field $[W = \oint \vec{E} \cdot d\vec{s}]$ $[\vec{E} \rightarrow \text{induced \& emf}]$.

$$W = \int \vec{F} \cdot d\vec{s} = (q_0 E)(2\pi R), \quad W = \oint \vec{F} \cdot d\vec{s} = q_0 \oint \vec{E} \cdot d\vec{s}$$

Substituting $\oint \vec{E} \cdot d\vec{s}$ for W

$$\oint \vec{E} \cdot d\vec{s} = \oint \vec{E} \cdot d\vec{s}$$

$$\boxed{-\frac{d\Phi_B}{dt} [\text{Faraday's Law}] = \oint \vec{E} \cdot d\vec{s}}$$

Consider a charge particle with q_0 moving around the circular path. Work done by induced electric field is $\oint \vec{E} \cdot d\vec{s}$.



This equation says simply that a changing magnetic field induces an electric field.

A new look at Electric Potential

Electric potential has meaning only for electric fields that are produced by static charges; it has no meaning for electric fields that are produced by induction.

$$V_f - V_i = - \int_i^f \vec{E} \cdot d\vec{s}$$

But Faraday's law, i and f are the same point (closed loop). The path connecting them is a closed loop, V_i and V_f are identical.

$$\oint \vec{E} \cdot d\vec{s} = 0.$$

However, when a changing magnetic flux is present, this integral is not zero but is $-\frac{d\Phi_B}{dt}$. Thus, assigning electric potential to an induced electric field leads us to a contradiction.

→ Electric potential has no meaning for electric fields associated with induction.

Inductors and Inductance

○ previously, we discussed that a capacitor can be used to produce a desired electric field.

○ Similarly, an inductor can be used to produce a desired magnetic field.

□ If we establish a current i in the turns of the solenoid we are taking as our inductor. The inductance of the inductor is then defined in terms of that flux as $\rightarrow$

$$L = \frac{N \Phi_B}{i} \quad (\text{induced defined}).$$

$N \rightarrow$ number of turns. The ^{turns} windings of the inductor are said to be linked by the shared flux.

The product $N \Phi_B$ is called the magnetic flux linkage

Inductance of a Solenoid

L is measure of the flux produced by the inductor per unit of current.

$\rightarrow$ A long solenoid of cross-sectional area A . What is the inductance per unit length near its middle?

$\rightarrow$ Consider a length l near the middle of this solenoid

flux there is $N \Phi_B = (nl)(BA)$ [$n \rightarrow$ number of turns per unit length of the solenoid]

$$B = \mu_0 i n,$$

$$\begin{aligned} L &= \frac{N \Phi_B}{i} = \frac{(nl)(BA)}{i} \\ &= \frac{(nl)(\mu_0 i n)(A)}{i} \\ &= \mu_0 n^2 l A \end{aligned}$$

$B \rightarrow$ magnitude of the magnetic field within the solenoid]

$$\frac{L}{l} = \mu_0 n^2 A \quad (\text{Solenoid})$$

Self-Induction

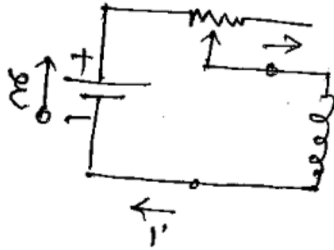
An induced emf $\mathcal{E}_L$ appears in any coil in which the current is changing. This process is called self-induction and the emf that appears is called a self-induced emf.

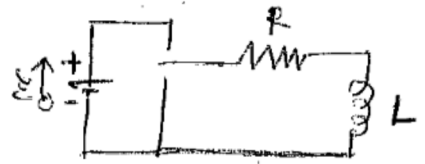
for any inductor

$$N\phi_B = Li$$

from Faraday law $\Rightarrow \mathcal{E}_L = - \frac{d(N\phi_B)}{dt}$

$$\mathcal{E}_L = -L \frac{di}{dt} \quad (\text{self induced emf}).$$



RL circuitsRC Circuit-

$q = C\mathcal{E}(1 - e^{-t/\tau_c})$ the rate at which the charge build up immediately to its final equilibrium value $C\mathcal{E}$ but approaches it in an exponential fashion.

→ rate of charge build up is determined by the capacitive time constant $\boxed{\tau_c = RC}$.

If we suddenly remove the emf from this same circuit, the charge does not immediately fall to zero but approaches zero in an exponential fashion.

$$q = q_0 e^{-t/\tau_c} \quad \left| \begin{array}{l} \tau_c \text{ (time constant) describes} \\ \text{the fall of the charge as} \\ \text{well as its rise} \end{array} \right.$$

Analogous slowing or rise of the current occurs if we introduce an emf $\mathcal{E}$ into a single-loop circuit containing a resistor R and an inductor L .

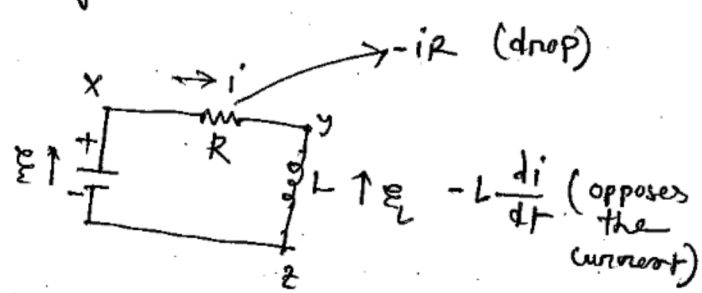
① If the inductor were not present, the current would rise rapidly to a steady value $\mathcal{E}/R$.

② Because of the inductor, a self-induced emf $\mathcal{E}_L$ appears in the circuit; → Lenz's law, this emf opposes the rise of the current, which means that it opposes the battery emf $\mathcal{E}$ in polarity.

Thus, the current in the resistor responds to the difference between two emfs, (i) A constant $\mathcal{E}$ due to the battery, (ii) A variable $\mathcal{E}_L (= -L \frac{di}{dt})$ due to the self induction.

As time goes on, the rate at which the current increases becomes less rapid and the magnitude of the self-induced emf, which is proportional to di/dt , becomes smaller. Thus, the current in the circuit approaches $\mathcal{E}/R$ asymptotically.

Initially, an inductor acts to oppose changes in the current through it. A long time later, it acts like an ordinary connecting wire.



Loop rules

$$-iR - L \frac{di}{dt} + \mathcal{E} = 0$$

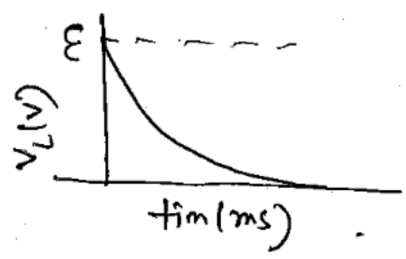
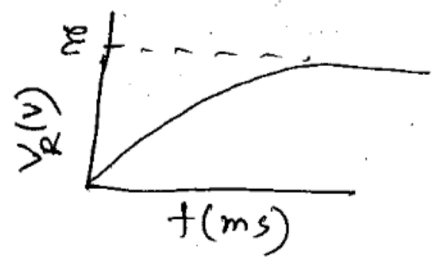
$$L \frac{di}{dt} + Ri = \mathcal{E} \quad \text{--- (1)}$$

Solution

$$i = \frac{\mathcal{E}}{R} (1 - e^{-Rt/L})$$

$$= \frac{\mathcal{E}}{R} (1 - e^{-t/\tau_L}) \rightarrow \boxed{\tau_L = \frac{L}{R}}$$

time constant



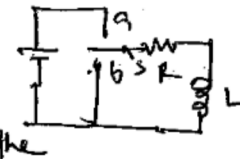
Time Constant:

$$i = \frac{\mathcal{E}}{R} (1 - e^{-t/\tau_L})$$

If we put $t = \tau_L = L/R$ in this equation, it reduces to

$$i = \frac{\mathcal{E}}{R} (1 - e^{-1}) = 0.63 \frac{\mathcal{E}}{R}$$

Thus, the time constant τ_L is the time it takes the current in the circuit to reach about 63% of its final equilibrium value $\mathcal{E}/R$.

Current Decay If the battery switch S is closed on a long enough for the equilibrium current $\mathcal{E}/R$ to be established and then is thrown to b, the effect will be to remove the battery from the circuit.  $\rightarrow$ with the battery gone, the current through the resistor will decrease. However, it can't drop immediately to zero but must decay to zero over time.

Now $\mathcal{E} = 0$,

$$L \frac{di}{dt} + iR = 0$$

$$\boxed{i = \frac{\mathcal{E}}{R} e^{-t/R} = i_0 e^{-t/\tau_L}} \quad | \quad i(0) = i_0 = \frac{\mathcal{E}}{R}$$

Decay of current.

i_0 is used to represent current at $t=0$.

In our case it is $\mathcal{E}/R$, but may be a different value.

Energy Stored in a Magnetic Field

⊛ When we pull two charged particles of opposite signs away from each other, we say that the resulting electric potential energy is stored in the electric field of the particles.

□ In the same way we say energy is stored in a magnetic field, but now we deal with current instead of electric charges.

$$\mathcal{E} = L \frac{di}{dt} + iR$$

[equation of current growth in this circuit]

$$\mathcal{E}i = Li \frac{di}{dt} + i^2 R \quad \text{--- (I)}$$

$$\frac{dU_B}{dt} = Li \frac{di}{dt} \quad \text{--- (II)}$$

The rate at which the battery does work ~~and~~ it is $\frac{(\mathcal{E}di)}{dt}$ work done.

→ Conservation of energy for RL Circuits, the middle term must represent the rate $\frac{dU_B}{dt}$ at which magnetic potential energy U_B is stored in the magnetic field. Thus,

$$\frac{dU_B}{dt} = Li \frac{di}{dt}$$

$$dU_B = Li di$$

$$\int_0^{U_B} dU_B = \int_0^i Li di$$

$$U_B = \frac{1}{2} Li^2 \quad (\text{magnetic energy})$$

Energy stored in electric field by a capacitor with

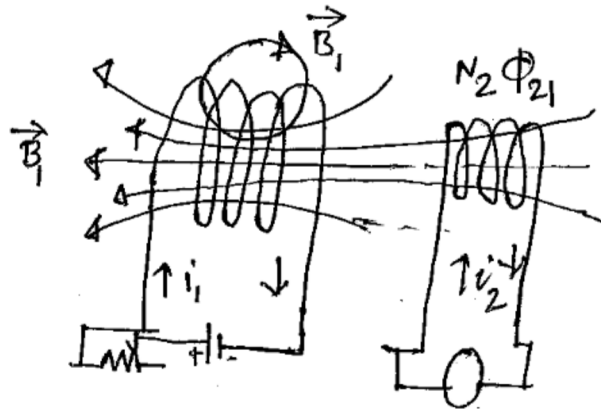
Capacitance $\boxed{U_E = \frac{q^2}{2C}}$

So, i^2 corresponds to q^2 .

~~Energy~~Mutual Induction

If we change i with time, an emf $\mathcal{E}$ given by Faraday's law appears in the second coil; we called this process induction. We called it mutual induction.

When one coil is used, we call it self-induction.



$$M_{21} = \frac{N_2 \Phi_{21}}{i_1} \quad \left[\text{similar to } L = \frac{N\Phi}{i} \right]$$

$$M_{21} i_1 = N_2 \Phi_{21}$$

If i_1 to vary with time by varying R , we have

$$N_{21} \frac{di_1}{dt} = N_2 \frac{d\Phi_{21}}{dt}$$

according to Faraday's law

$$\mathcal{E}_2 = -M_{21} \frac{di_1}{dt} \quad \left(\mathcal{E} = -L \frac{di}{dt} \right)$$

Now, if we vary i_2 by varying R , we then have, by the argument given above,

$$\mathcal{E}_1 = -M_{12} \frac{di_2}{dt} \quad \text{--- (1)}$$

Chapter 31
Electromagnetic oscillations and

Alternating current.

Q In an oscillating LC circuit, energy is shuttled periodically between the electric field of the capacitor and the magnetic field of the inductor.

Instantaneous values of the two forms of energy are,

$$U_E = \frac{q^2}{2C} \quad \text{and} \quad U_B = \frac{Li^2}{2}$$

Where q is the instantaneous charge on the capacitor and i is the instantaneous current through the inductor.

Q $\rightarrow$ The total energy $U (= U_E + U_B)$ remains constant.

Q $\rightarrow$ The principle of conservation of energy leads to

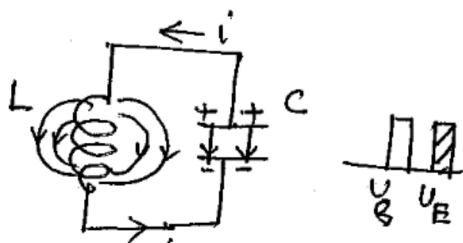
$$L \frac{d^2 q}{dt^2} + \frac{1}{C} q = 0 \quad (\text{LC oscillations})$$

Solution: $q = Q \cos(\omega t + \phi)$ (charge) | Q is the charge amplitude

$i = -\omega Q \sin(\omega t + \phi)$ (current)

$\omega \rightarrow \frac{1}{\sqrt{LC}}$

$\omega Q = I$, is the current amplitude



The block - Spring oscillation.

$$U = U_b + U_s = \frac{1}{2}mv^2 + \frac{1}{2}kx^2 \quad [U - \text{total energy}]$$

↓
kinetic energy of the moving block
→ potential energy of the stretched string.

If there is no friction, U remains constant over time, even though v and x vary. $\frac{dU}{dt} = 0$.

$$\frac{dU}{dt} = mv \frac{dv}{dt} + kx \frac{dx}{dt} = 0$$

$$v = \frac{dx}{dt}, \quad \frac{dv}{dt} = \frac{d^2x}{dt^2} \quad \text{we find}$$

$$m \frac{d^2x}{dt^2} + kx = 0 \quad (\text{block-spring oscillation})$$

$$x = X \cos(\omega t + \phi) \quad (\text{displacement})$$

Similarly

$$U = U_B + U_E = \frac{L i^2}{2} + \frac{q^2}{2C}$$

$$\frac{dU}{dt} \Rightarrow L i \frac{di}{dt} + \frac{q}{C} \frac{dq}{dt} = 0$$

$$L \frac{dq}{dt} + \frac{1}{C} q = 0 \quad (LC \text{ oscillations})$$

$$q = Q \cos(\omega t + \phi) \quad \text{--- (I)}$$

$$i = \frac{dq}{dt} = -\omega Q \sin(\omega t + \phi) \quad \text{--- (II)} \\ = -I \sin(\omega t + \phi)$$

$$\frac{dq}{dt} = -\omega^L Q \sin(\omega t + \phi) \quad \downarrow$$

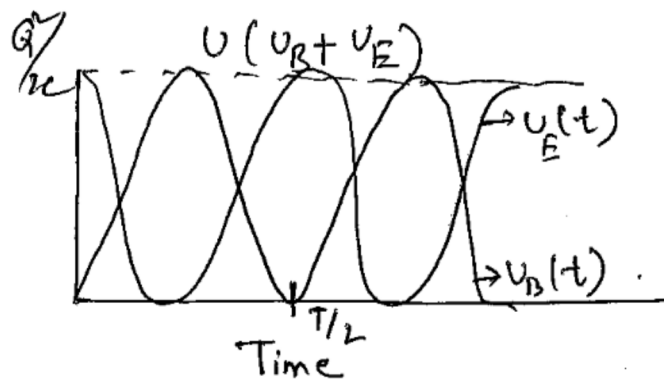
$$L \frac{dq}{dt} + \frac{1}{C} q = 0$$

$$-L \omega^L Q \cos(\omega t + \phi) + \frac{1}{C} Q \cos(\omega t + \phi) = 0$$

$$\omega = \frac{1}{\sqrt{LC}} \quad - (iii)$$

$$U_E = \frac{q^L}{2C} = \frac{Q^L}{2C} \cos^2(\omega t + \phi)$$

$$U_B = \frac{1}{2} L i^L = \frac{1}{2} L \omega^L Q^L \sin^2(\omega t + \phi)$$



The stored magnetic energy and electric energy in the circuit.